Yusuf Ahmad • US Citizen

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OBJECTIVE

Seeking an RTL/SoC design engineering role applying hands-on experience in Verilog/SystemVerilog RTL design, UVM verification, AXI-based peripherals, CDC-safe architectures, cryptographic accelerators, and pipelined RISC-V CPU development.

EDUCATION

Bachelor of Technology Amity University Uttar Pradesh Lucknow | 2022 – 2026

Electronics & Communication Engineering

- CGPA: 8.66 *(Core Subjects CGPA - 3.80 / 4.00 approx.)*
- Thesis: " Evaluation of RISC-V System-on-Chip (SoC) Interface Architecture"
  Advisor: Dr. Kamlesh Kumar Singh | Focus: Memory-mapped integration of a RISC-V core with a custom ChaCha20 cryptographic accelerator.

PUBLICATIONS

Hardware Implementation of ChaCha20 Accelerator for RISC-V System — IEEE IC3ECSBHI, 2026 | Published

Low Power Encryption for IoT Devices using FPGA and Microcontroller — IEEE CNC, 2025 | Published

EXPERIENCE

Ardent Computech Pvt. Ltd. - VLSI IC Design Intern **May - July 2025**

Kolkata, West Bengal

- Independently led the end-to-end design and verification of a lightweight FPGA-based encryption module (Low-Power LFSR XOR Encryption Core), implementing an SPI-driven XOR cipher with dynamically evolving LFSR key scheduling for secure sensor data transmission.
- Developed synthesizable VHDL RTL for serial-to-parallel SPI data capture, dual-flip-flop clock synchronization for metastability mitigation, and cycle-accurate byte-level encryption; verified functionality using custom self-checking testbench in Vivado simulation environment.
- Achieved post-implementation power estimates of ~26–35 mW with <1% LUT utilization (75 LUTs, 48 FFs) under Spartan-7 constraints, enabling single-cycle encryption latency per 8-bit payload and SPI throughput up to 10 MHz for real-time embedded communication workloads.

PROJECTS

Custom RISC-V SoC with ChaCha20 Hardware Accelerator *(SystemVerilog, Vivado)* **January 2026 - February 2026**

- Developed a custom RISC-V–based SoC with modular CPU datapath, control logic, ALU, load/store unit, and branch handling.
- Integrated a memory-mapped ChaCha20 cryptographic accelerator for hardware-accelerated encryption within the SoC memory space.
- Built cycle-accurate simulations and modular assembly test programs to validate instruction execution, memory operations, and CPU–accelerator interaction.

Pipelined ChaCha20 Encryptor with RISC-V Interface *(Sys.Verilog, Tcl, Python, Batch script, Vivado)* **Dec 2025 - Jan 2026**

- Designed modular ChaCha20 engine (ARX quarter-round pipeline, key scheduler, FSM control) with a 32-bit memory-mapped interface for RISC-V systems.
- Verified using SystemVerilog testbench + Python reference vectors; matched outputs to software model across 20-round configurations.
- Achieved timing closure at 100 MHz on Spartan-7: ~799 LUTs, ~741 FFs, ≈0.067 W total power; sustained ~300 MB/s post-synthesis throughput.

AXI4-Lite Slave Peripheral *(Verilog, ModelSim, Tcl Automation)* **August 2025 - February 2026**

- Architected a modular AXI4-Lite slave with independent AW/W/B/AR/R channel FSMs, byte-enable (WSTRB) masking, and protocol-compliant OKAY/SLVERR/DECERR response generation.
- Built a self-checking constrained-random verification environment (driver, monitors, scoreboard) and validated design correctness over 300K+ AXI transactions with zero data mismatches or deadlocks.

- Implemented transaction tracking and response-aware checking to verify read-after-write integrity and protocol error handling under interleaved traffic, producing regression summaries and waveform-based validation artifacts.

CDC-Safe Asynchronous FIFO with UVM Verification *(SystemVerilog, UVM, QuestaSim)* **July 2025 - March 2026**

- Designed dual-clock asynchronous FIFO using Gray-coded pointer synchronization and two-flip-flop CDC synchronizers to safely transfer data between independent clock domains while mitigating metastability.
- Developed a transaction-level UVM verification environment with read/write agents, sequencers, drivers, monitors, functional coverage collectors, and a self-checking scoreboard for FIFO ordering validation.
- Executed randomized concurrent traffic with 900 write and 897 read transactions, achieving 100% data integrity through queue-based scoreboard comparison with zero ordering mismatches.

TECHNICAL SKILLS

RTL Design & Digital Architecture:

Asynchronous FIFO design, CDC-safe architectures, Gray-coded pointer synchronization, metastability mitigation, FSM design, pipelined data paths, cryptographic hardware accelerators (ChaCha20), dual-port memory systems.

Protocols & Interfaces:

AXI4-Lite, SPI (with CDC synchronization), memory-mapped peripheral interfaces, register-mapped hardware control.

Verification & Methodology:

UVM-based verification (agents, sequencers, drivers, monitors, scoreboard, analysis ports), constrained-random verification, transaction-level testbenches, protocol compliance testing, waveform debugging, SystemVerilog Assertions (SVA – theoretical understanding).

EDA & Simulation Tools:

Siemens QuestaSim, ModelSim, Xilinx Vivado Design Suite, EDA Playground.

FPGA Development:

Xilinx Spartan-7 FPGA design flow, synthesis, implementation, timing analysis, power estimation, utilization analysis, RTL schematic exploration.

Programming & Scripting:

C++, Python, Tcl scripting, Makefiles, simulation automation, regression scripting, randomized test generation.

Hardware & Design Tools:

MATLAB, LTSpice (analog simulation), MicroWind (CMOS layout experimentation), EasyEDA (schematic capture & PCB design), GitHub version control.